

1.5 Solve problems on a calculator using mathematical calculations for:

1.5.1 Real numbers

1.5.2 Exponents

1.5.3 Percentages

Real Numbers

Real numbers ($\mathbb{R}$) are all the numbers that can be placed on a number line. They include both rational and irrational numbers.

Number System	Symbol	Meaning	Examples
Natural numbers	$\mathbb{N}$	Counting numbers	1, 2, 3, 4, ...
Whole numbers		Natural numbers plus zero	0, 1, 2, 3, ...
Integers	$\mathbb{Z}$	Whole numbers and their negatives	...-2, -1, 0, 1, 2...
Rational numbers	$\mathbb{Q}$	Numbers that can be written as a fraction of two integers	$\frac{3}{4}$, -2, 0.6, 1.25
Irrational numbers		Numbers that cannot be written as a fraction of two integers; their decimals do not terminate or repeat	$\sqrt{2}$, $\sqrt{5}$, π

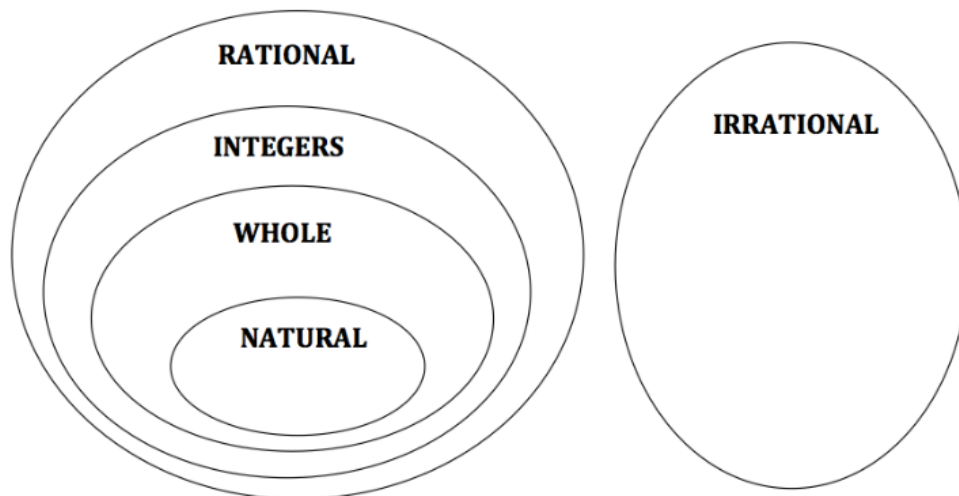
The number systems overlap.

For example,

5 is a natural number, whole number, integer, rational number, and real number.

-3 is an integer, rational number, and real number.

Pi is irrational and real.



To which number systems do the following belong? Check all that apply.

Number	Natural	Whole	Integer	Rational	Irrational	Real
3.19						
$-\frac{3}{8}$						
0						
-10						
0.272727...						
$\sqrt{17}$						

Real Number Calculations

Evaluate. Use the order of operations.

a) $-8.4 + 15.7$

b) 7.25×4.6

c) $20 - 4.5 \times 3$

d) $(20 - 4.5) \times 3$

e) $-12.6 + 8.9$

f) $45.6 \div 3.8$

g) $18 - 2.5 \times 4$

h) $(18 - 2.5) \times 4.$

Exponents

The Power Rule (extension)

To raise a product to a power, each factor inside the parentheses must be raised to the power.

$$(xy)^a = x^a y^a$$

To raise a quotient to a power, the numerator and denominator inside the parentheses must be raised to the power.

$$\left(\frac{x}{y}\right)^a = \frac{x^a}{y^a}$$

Examples

a) $(5x^2)^3$

b) $(2x^3)^4$

c) $\left(\frac{2x}{3}\right)^2$

d) $(3x)^2 (2x^3)^4$

e) $(3x^{-1}) (12x)^{-1}$

f) $\left(\frac{4}{x}\right)^2 \left(\frac{2}{x}\right)^{-3}$

g) $(a^{-6}b)^{-1}(a^3b^{-1})^{-2}$